TIC 201429480

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period       2.892472 d
  Epoch (t0)   1469.0627 BTJD
  Duration     0.1200 d
  Depth        10.871 %  (108,713 ppm)
  SNR          1625.6

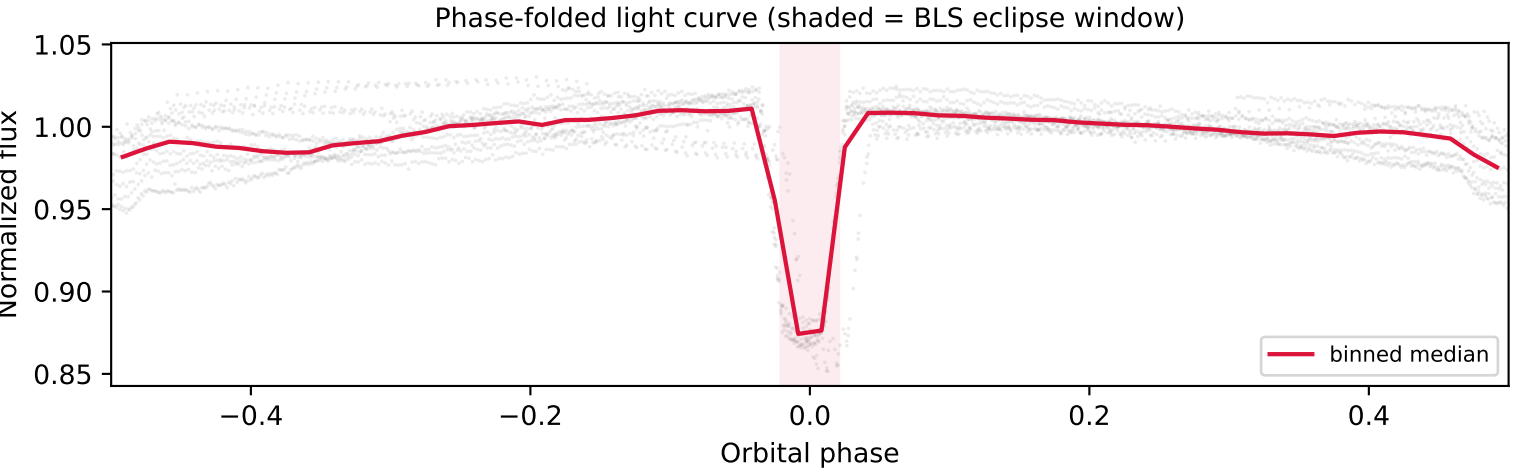
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     15  (gate >= 3)
  Robust depth ratio    1.13  (gate >= 0.50)
  Robust depth sigma    103.5  (gate >= 7.0)
  Odd/even z-score      2.88  (consistent)
  Secondary eclipse     not detected
```

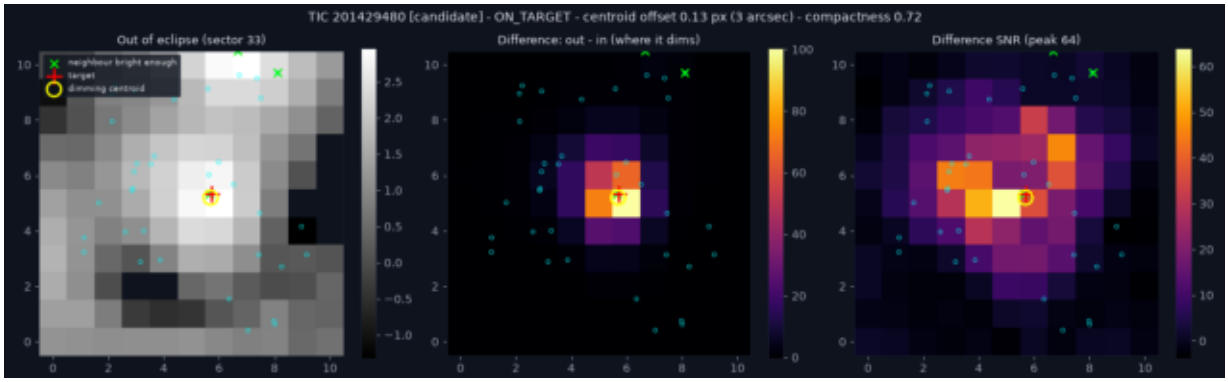
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 33    light-curve dip in this sector: 113.0 sigma    t0 refined by -0.0050 phase
  Difference-image peak SNR   64.00   (gate >= 5.0)      Compactness   0.72   (gate >= 0.50)
  Dimming centroid offset     0.13 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 705926341
    0.13 px away, dTmag +2.06 - farther from the dimming than the target
```





Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.001 px (dimming centroid 0.134 px from target vs 0.134 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.